Supplemental Material for
"Reading a magnet's Hamiltonian term by term: two-dimensional terahertz spectroscopy as vertex spectroscopy of $TmFeO_3$"

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Contents

S1 The SU(3) qutrit and why its classical dynamics is quantum-exact

Tm^{3+} is $4f^{12}$, $^3H_6$ ($J = 6$). In the $C_s(m)$ crystal field of the *Pbnm* $4c$ site the 13-fold multiplet splits into 13 singlets; the model keeps the three lowest, $|1\rangle, |2\rangle, |3\rangle$, with splittings $E_{12} = 2.068\,\mathrm{meV}$ ($0.50\,\mathrm{THz}$) and $E_{13} = 4.963\,\mathrm{meV}$ ($1.20\,\mathrm{THz}$), consistent with the measured lines $E_{12} + E_{23} \approx E_{13}$. The site state is the Gell-Mann vector $\lambda^a = \mathrm{Tr}(\rho\Lambda^a)$, $a = 1 \ldots 8$, and every single-ion term (CEF energies, external drives, exchange mean fields) is *linear* in the Λ^a. For a Hamiltonian linear in the generators the von Neumann equation $\dot\rho = -i[H,\rho]$ closes on $\vec\lambda$:

$$\dot\lambda^a \;=\; \Omega_{ab}(t)\,\lambda^b - \gamma_a\big(\lambda^a - \lambda^a_{\rm eq}\big), \qquad \Omega_{ab} = \tfrac{1}{2}\,\mathrm{Tr}\Big(\Lambda^a\,(-i)[H(t),\Lambda^b]\Big)\,. \tag{1}$$

The classical SU(3) propagation is therefore the *exact* quantum dynamics of the driven, damped three-level ion, and a Boltzmann-weighted ensemble of trajectories launched from the level seeds $|1\rangle, |2\rangle, |3\rangle$ is the exact thermal density matrix for single-ion drives. (This was verified explicitly against an independent density-matrix integrator; the two censuses agree to all digits quoted.) Damping is Bloch-type on the Gell-Mann pairs of each transition, $\gamma(\lambda^{1,2}), \gamma(\lambda^{4,5}), \gamma(\lambda^{6,7})$, with population relaxation $\gamma(\lambda^{3,8})$ toward the equilibrium populations captured at initialization.

The magnetically bright coordinate of the E_{12} block follows from the projected moment $PJ_zP = 5.264\,\lambda^2$: only λ^2 radiates magnetically, and the onsite splitting drives the planar precession $\dot\lambda^1 = -\Delta_{12}\lambda^2$, $\dot\lambda^2 = +\Delta_{12}\lambda^1$, so the E_{12} sector is a single driven oscillator

$$\lambda^2(t) = \int_0^t \sin[\omega_{12}(t-t')]\,h^1(t')\,dt', \tag{2}$$

where h^1 is the effective force a conversion vertex exerts on λ^1. All delay-axis structure of a cross peak lives in the τ-dependence of h^1.

S2 Crystallography and the projected Fe–Tm coupling

$TmFeO_3$ is $Pbnm$ ($Z=4$): Fe^{3+} at $4b$ (site symmetry $\bar 1$; every Fe sits on an inversion center), Tm^{3+} at $4c$ (site symmetry the single mirror m_z). Inversion maps each Fe sublattice to itself and exchanges Tm sublattices in pairs; the local mirror is the physical $4c$ site mirror and generates the relative λ^5, λ^7 signs between Tm sublattices. The Fe order is G-type ($T_N \approx 632\,\mathrm{K}$) with DM canting; below the spin reorientation ($T_2 \approx 85\,\mathrm{K}$) the ground state is $\Gamma_2 = (F_x, C_y, G_z)$.

The Fe–Tm sector is built from the physical operator route

$$\boldsymbol{S},\ \boldsymbol{J} \;\longrightarrow\; H_{\rm Fe-Tm} = \boldsymbol{S}^T A\,\boldsymbol{J} \;\longrightarrow\; P\boldsymbol{J}P \;\longrightarrow\; \text{couplings to } \lambda^a, \tag{3}$$

never by assuming that a space-group rotation acts as a closed onsite SU(3) rotation in one fixed qutrit basis. Working in a *transported gauge* (qutrit bases on the four Tm sublattices related by the group operations themselves) makes every bond formula explicit and turns the irrep (Γ_1) projection into simple four-site sign sums. Time-reversal splits the Gell-Mann set into $\lambda^- = \{\lambda^2, \lambda^5, \lambda^7\}$ (odd) and $\lambda^+ = \{\lambda^1, \lambda^3, \lambda^4, \lambda^6, \lambda^8\}$ (even); every coupling term must be $\mathcal{T}$-even overall, which fixes how many Fe-spin or B-field legs may attach to each sector. The candidate couplings through cubic order in the fields are

family	operator form	sector	legs
K^-	$S_\alpha\lambda^a$	λ^-	1 Fe
ESJ	$E_\eta S_\alpha \lambda^a$	λ^-	1 E-photon + 1 Fe
κ^B	$B_\eta S_\alpha \lambda^a$	λ^+	1 B-photon + 1 Fe
W	$S_\alpha S_\beta \lambda^a$	λ^+	2 Fe

The full orbit formulas (all eight coefficients per bond, all four Tm sublattices, both time-reversal sectors) and the implementation prescriptions for global-axis versus local-frame storage are given in the long-form working note that accompanies the data (`tmfeo3_foundation.tex`, Secs. 3–14).

S3 The sharp requirement and the vertex scorecard

Two collinear pulses give $M_{\mathrm{NL}} = M_{AB} - M_A - M_B$; order by order only the mixed terms survive, so M_{NL} starts at second order and the leading robust magnetic-dipole contribution is third order. Combining Eq. (2) with the conservation rules at each vertex (energy; $\mathcal{T}$-parity; Γ_1) yields six requirements for a coupling to produce the geometry-A cross peak at $(q_{\mathrm{AFM}}, E_{12})$:

R1 (detection) reaches λ^1/λ^2 (the E_{12} block);

R2 (drive) is sourced by the pumped δS_y magnon;

R3 (parity) is $\mathcal{T}$-even (fixes the $\lambda^{\pm}$ sector);

R4 (irrep) is Γ_1-allowed in the Γ_2 state;

R5 (energy) carries the $0.90 \to 0.50\,\mathrm{THz}$ mismatch (needs ≥ 2 non-λ legs);

R6 (M_{NL}) involves both pulses.

Table S1: Vertex scorecard. Only W and κ^B pass all six requirements.

vertex	legs	sector	R1	R2–R6	verdict
K^-	$1\mathrm{Fe} + 1\mathrm{Tm}^-$	λ^-	$\times$	$\times$	hybridizes only; Γ_1-blocked; reorients Γ_2
ESJ	$E + \mathrm{Fe} + \mathrm{Tm}^-$	λ^-	$\times$	✓	real, but feeds E_{13}/E_{23}, not E_{12}
W	$2\mathrm{Fe} + \mathrm{Tm}^+$	λ^+	✓	✓	**winner**: rank-1, factorized peak
κ^B	$B + \mathrm{Fe} + \mathrm{Tm}^+$	λ^+	✓	✓	works; B_z-gated $\Rightarrow$ geometry-B vertex

K^- fails threefold. (i) Linear in S, it only rotates the harmonic normal-mode basis: coupled harmonic oscillators give diagonal peaks only; a static bilinear conserves energy and cannot convert 0.90 into 0.50 THz. (ii) The Γ_1 orbit sums vanish for every channel except K^-_{2y}: a static one-at-a-time scan at 0.2 meV leaves the energy, the Fe order, and the uniform $(\lambda^2, \lambda^5, \lambda^7)$ *bit-identical* to baseline for all components except $2y$. (iii) The one allowed channel K^-_{2y} is a transverse molecular field on the order parameter and reorients $\Gamma_2 \to G_y$ between 0.085 and 0.090 meV—it is not a perturbative vertex. Full-dynamics 2DCS on the inert channels (K^-_{2x} up to 1.5 meV, K^-_{2z} up to 2.0 meV) leaves the induced λ^2 at machine precision ($\leq 7 \times 10^{-15}$).

The two winners. The two-magnon vertex rectifies the pulse-A/pulse-B cross term,

$$h^1_{\mathrm{NL}}(t,\tau) = W_1^{yy} A^2 \left[\cos(q_{\mathrm{AFM}}\tau) - \cos(q_{\mathrm{AFM}}(2t+\tau))\right] \theta(t) \;\Rightarrow\; M^W_{\mathrm{NL}} \propto \cos(q_{\mathrm{AFM}}\tau)\left[1 - \cos\omega_{12} t\right], \quad (4)$$

a clean factorized peak with no diagonal partner. The field-gated vertex kicks at the probe, $h^1 \propto B_y(t) S_y(t)$ with $S_y(0) = A \sin q_{\mathrm{AFM}}\tau$, giving $M^\kappa_{\mathrm{NL}} \propto \sin(q_{\mathrm{AFM}}\tau)[1 - \cos\omega_{12} t]$: also on target, but the same gate dumps each pulse's local response into Tm at both pulse times, generating an (E_{12}, E_{12}) diagonal and a $(0, E_{12})$ background. Its dominant symmetry slice is z-gated ($B_z(t)$), so in geometry A ($B_z = 0$) it is *identically silent*: the gate is simultaneously the reason it is harmless in geometry A and active in geometry B.

Exhaustion of the quadratic families. All remaining symmetry-allowed quadratic couplings were switched on one at a time in the full nonlinear dynamics and are exactly inert (differences at the 10^{-12} level or below): $W_4^{yy,xy,yz,xx,zz,xz}$, W_1^{xy}, $W_6^{yy,xz}$, κ_{2y}^{E}, κ_{5x}^{E}. Among the W_1 components: W_1^{yz} inert, W_1^{zz} negligible, W_1^{xx} alive but floods the spectrum with an $\omega_t = 0.16\,\mathrm{THz}$ comb, W_1^{xz} alive and on-target for hybridization (used below), and no static component can replace the gated κ^B for the geometry-B transfer peak: the entire W_1+W_4 family fails to produce $(E_{12}, q_{\mathrm{AFM}})$ in geometry B without simultaneously contaminating geometry A. The B_z-gate is therefore *proven necessary by exhaustion.*

S4 Numerical protocol

Dynamics. Classical LLG for Fe (unit spins, $S = 5/2$ scale absorbed), SU(3) Bloch dynamics for the four Tm sublattices; RK4 with $\Delta t = 0.02$ code units; $1 \times 1 \times 1$ cell of the $Pbnm$ magnetic unit (8 Fe + 4 Tm). Time is measured in $\hbar$/meV; a frequency f in THz corresponds to $E = f \times 4.136\,\mathrm{meV}$.

2DCS protocol. Delay scan $\tau \in [-120, 0]$ code units, $\Delta\tau = 0.1$ (0.2 for scans); detection window to $t_{\max} = 160$ for 0.01 THz resolution on ω_t. The nonlinear signal is the honest three-run subtraction $M_{\mathrm{NL}} = M_{AB} - M_A - M_B$ per delay; no pulse-window chunking; no reuse of the reference run across delays (both shortcuts were found to contaminate spectra and are documented pitfalls). The pump enters as a tabulated waveform (the measured field, times converted $t_{\mathrm{code}} = t_{\mathrm{ps}}/0.6582$, auto-centered and normalized).

Spectra and blind peak census. 2D spectra are $|\mathrm{FFT}_{\tau,t}|$ of M_{NL} with: detection window $t \geq 3$ (excludes pulse overlap), Gaussian t-apodization to 0.03, cosine tapers on both τ edges, global mean subtraction. The delay axis is plotted physically ($\omega_T = -\omega_\tau^{\mathrm{FFT}}$; $+ =$ non-rephasing). Peaks are accepted only by a blind census: 2D local maxima (maximum filter, 0.10 THz neighborhood) with prominence ≥ 1.5 over the local median background (0.34 THz annulus) and amplitude $\geq$ 4–5% of the map maximum, merged within 0.09 THz. Rectified ($\omega_t \approx 0$) content is analyzed separately via $S_{\mathrm{DC}}(\tau) = \langle M_{\mathrm{NL}} \rangle_t$ with cubic drift removal.

Thermal ensembles. Boltzmann mixtures of trajectories from the level seeds $|1\rangle, |2\rangle, |3\rangle$ (Sec. S1). Two documented pitfalls: (i) any pre-dynamics energy minimization collapses excited-level seeds and must be disabled; (ii) population damping $\gamma(\lambda^{3,8})$ relaxes toward the equilibrium captured at initialization, which must therefore be captured per-seed.

S5 Experimental inputs

Two consequences organize the readout physics. First, excitation scales as E^2 and detection as E^1 (third order). Second, the CEF lines are E1-bright while q_{AFM} is E1-dark but M1-bright: a magnon is therefore read out either by borrowing a CEF electric dipole through an Fe–Tm vertex (geometry A) or through its own magnetic dipole at the magnon frequency (geometry B). Using the magnetic g-factor instead of $\sqrt{f}$ for the magnon electric emission over-weights it by $\sim 100\times$ and buries the CEF peaks under the magnon diagonal; the oscillator-strength weighting resolves this without any tuning.

Table S2: Measured mode parameters and the derived simulation inputs. CEF damping is Bloch on the Gell-Mann pair of each transition, $\gamma_a[\text{meV}] = \gamma[\text{THz}] \times h$; magnon damping is Gilbert, $\alpha = \gamma/2\nu$ fixed by the q_{AFM} line. Emission weight is $\sqrt{f}$ (E1) for the CEF lines and M1 for the magnons. The last column is the measured pulse spectral amplitude at each mode.

mode	ν [THz]	f	$\sqrt{f}$	γ [THz]	damping (sim)	pulse amp.
q_{FM}	0.38	0.08	0.28	0.005	$\alpha = 1.7 \times 10^{-3}$	0.92
q_{AFM}	0.90	7×10^{-4}	0.026	0.003	$\alpha = 1.7 \times 10^{-3}$	0.68
E_{12}	0.50	6.5	2.55	0.038	$\gamma(\lambda^{1,2}) = 0.157\,\text{meV}$	1.00
E_{23}	0.70	6.6	2.57	0.10	$\gamma(\lambda^{6,7}) = 0.414\,\text{meV}$	0.85
E_{13}	1.20	8.7	2.95	0.05	$\gamma(\lambda^{4,5}) = 0.207\,\text{meV}$	0.36

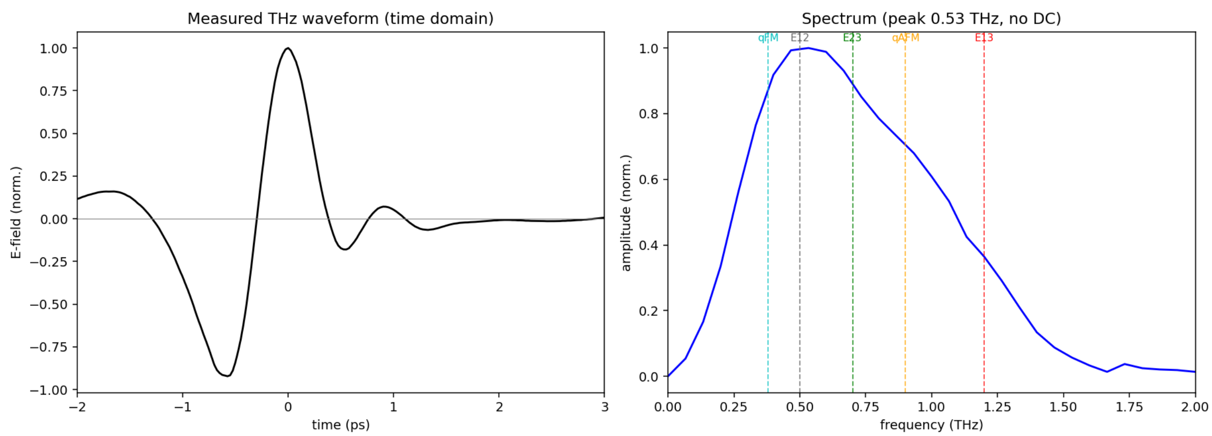

Figure S1: The measured THz pump. Left: single bipolar cycle ($\sim 1.2\,\text{ps}$, no DC). Right: amplitude spectrum peaked at $0.53\,\text{THz} \approx E_{12}$, covering all five modes. The absence of DC content removes the quasistatic-tilt legs that a Gaussian stand-in pulse would add; the measured waveform is the cleanest configuration.

S6 Displacive verification of the W mechanism

The rectified two-magnon force is a suddenly switched step: its edge carries spectral weight $\propto e^{-\omega_{12}^2\sigma^2/2}$ at the CEF frequency, i.e. the E_{12} quantum is drawn from the pulse bandwidth (intra-pulse difference-frequency/impulsive-Raman), while the magnon pair carries only the phase memory $\cos q_{\text{AFM}}\tau$. Verified directly: stretching the pulse $\sigma = 0.25 \to 1.0$ at fixed amplitude collapses the peak by 4.5 decades ($51.7 \to 1.7 \times 10^{-3}$), while switching the direct Tm drive on/off changes it by $< 10^{-4}$ (pure-W origin). The same sweep resolves the second Fe leg: impulsive pulses use the resonant probe magnon (peak $\propto A^2$); longer pulses cross over to the field-following quasistatic tilt (peak $\propto A^1$). The measured no-DC waveform suppresses the quasistatic legs, which is why the experimental pulse yields the cleanest spectra.

S7 Geometry B: gate versus buildup

Isolated-vertex runs: κ_{1y}^B alone $\Rightarrow (E_{12}, q_{\text{AFM}})$ present; W_1^{xz} alone $\Rightarrow$ peak absent. In the consolidated Hamiltonian $W_1^{yy} = W_1^{xz} = 0.01$ and $\kappa_{1y}^B = 0.0035$: the geometry-B peak ratio is the observable that calibrates κ^B, and because κ^B is B_z-gated the recalibration provably cannot affect either $H \parallel a$ channel (verified bit-identical).

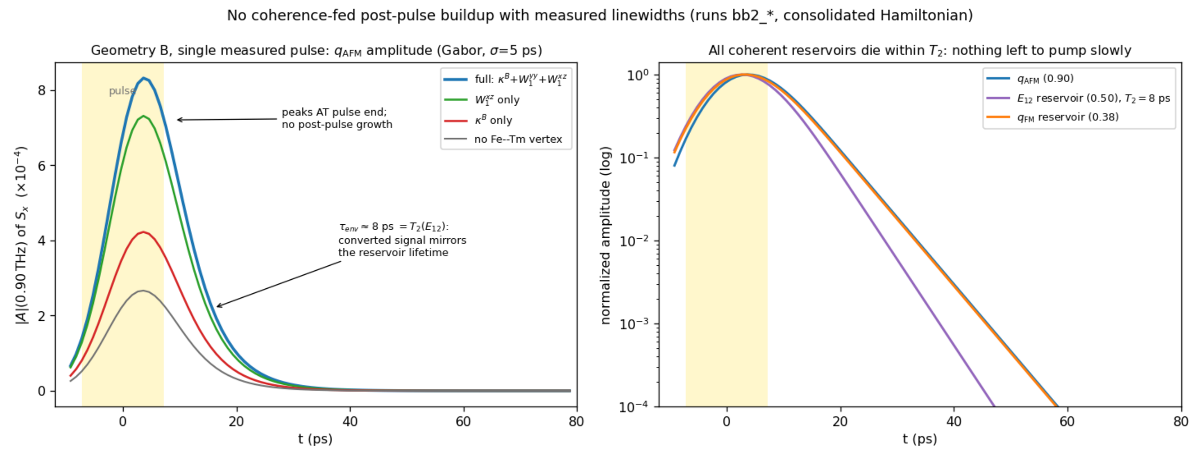

Figure S2: No coherence-fed post-pulse buildup once the measured linewidths are enforced (single measured pulse, geometry B, runs `bb2_*`; Gabor amplitude, $\sigma = 5\,\mathrm{ps}$, artifact-free local analysis). Left: the q_{AFM} amplitude for the four vertex configurations peaks *at* pulse end and decays with $\tau_{\mathrm{env}} \approx 8\,\mathrm{ps} = T_2(E_{12})$; the vertices set the absolute scale but none produces post-pulse growth. Right: all coherent reservoirs die within their measured T_2 (log scale)—there is nothing left to pump the magnon slowly.

The post-drive buildup: a corrected analysis. An earlier stage of this project attributed the experimentally reported post-pulse growth of the geometry-B signal to slow W_1^{xz}-mediated exchange through the nearly closed triad $q_{\mathrm{FM}} + E_{12} = 0.88 \approx q_{\mathrm{AFM}}$. That attribution does not survive the enforcement of the measured linewidths. Because $B \parallel c$ never drives q_{AFM} directly, *all* geometry-B magnon amplitude is converted, and a converted signal inherits its source's lifetime: with $T_2(E_{12}) = 8\,\mathrm{ps}$ the E_{12}-fed component (and with it the triad) is dead within $\sim 10\,\mathrm{ps}$. The artifact-free Gabor analysis of Fig. S2 shows the q_{AFM} amplitude peaking at pulse end and decaying monotonically in every vertex configuration; the earlier "buildup" was visible only while $\lambda^{1,2}$ were left undamped (an infinitely lived E_{12} reservoir, inconsistent with the measured $0.038\,\mathrm{THz}$ linewidth). W_1^{xz} therefore loses its buildup role—but it acquires a direct observable of its own instead (next paragraph); it remains bounded above ($\lesssim 0.02$) by the mixer sweep of the preceding paragraph.

The reverse transfer peak: W_1^{xz}'s own observable. The experiment reports an additional geometry-A *cross*-channel peak near (0.5, 1.0). The model contains a genuine local maximum at exactly $(+0.49, 0.90) = (E_{12}, q_{\mathrm{AFM}})$ in the λ^2 channel (0.063 of the main peak; run `vA`), with a weaker $2E_{12} = 1.00\,\mathrm{THz}$ harmonic companion at 0.021. Isolated-vertex tests identify the mechanism: with $W_1^{xz} \to 0$ the peak collapses sevenfold ($0.063 \to 0.009$; run `vA_noWxz`) while the main $(q_{\mathrm{AFM}}, E_{12})$ peak is untouched, and doubling the Tm drive doubles it ($0.063 \to 0.123$; run `vA_su3x2`), i.e. it is linear in the stored E_{12} amplitude. This is the *reverse* of the main transfer: the pump-stored E_{12} coherence is converted *into* the q_{AFM} magnon by the static hybridization—the static W family read in both directions, magnon$\to$CEF through W_1^{yy} and CEF$\to$magnon through W_1^{xz}. The measured amplitude ratio of the reverse to the forward peak calibrates W_1^{xz} directly. Because the reverse peak emits at the magnon frequency it is detected through the magnon's own M1 dipole; two further observed facts calibrate the detection and drive dials. (i) The $(q_{\mathrm{AFM}}, q_{\mathrm{AFM}})$ magnon diagonal is *not* observed: since the diagonal scales as the cube of the Zeeman amplitude while the main peak is quadratic and the reverse peak linear (verified: $3.49 \to 0.54 \to 0.12$, $3.54 \to 0.83 \to 0.20$, $1.16 \to 0.55 \to 0.27$ for $A_{\mathrm{Fe}} = 0.1/0.05/0.025$), its absence pins the Zeeman drive far below the E1 drive; the operating point $A_{\text{Fe}} = 0.010$, A_{Tm} reduced $\times 0.45$ (runs `vA10`) also suppresses the second-order-pump E_{13}-delay intruder at $(1.19, 0.50)$ to 0.28. (ii) The observed intensity *parity* of the two transfer peaks fixes the M1/E1 detection weight at $w \simeq 0.7$ (composite census in `data/census_crosspol_composite.json`). Final detected inventory: reverse $1.00 \approx$ main 0.94 co-dominant, E_{12} diagonal 0.93 (rephasing side), 1.4 THz combination tones 0.42–0.43 (standing prediction), rephasing partners 0.28–0.36, magnon diagonal below 0.28. Assignment discriminators for the experimental peak: an (E_{12}, q_{AFM}) reading emits exactly at the magnon frequency (0.86 THz measured), inherits the narrow magnon linewidth along ω_t, and softens toward the spin reorientation; the $2E_{12}$ harmonic reading sits rigidly at 1.00 THz, is broader, and grows one field power faster.

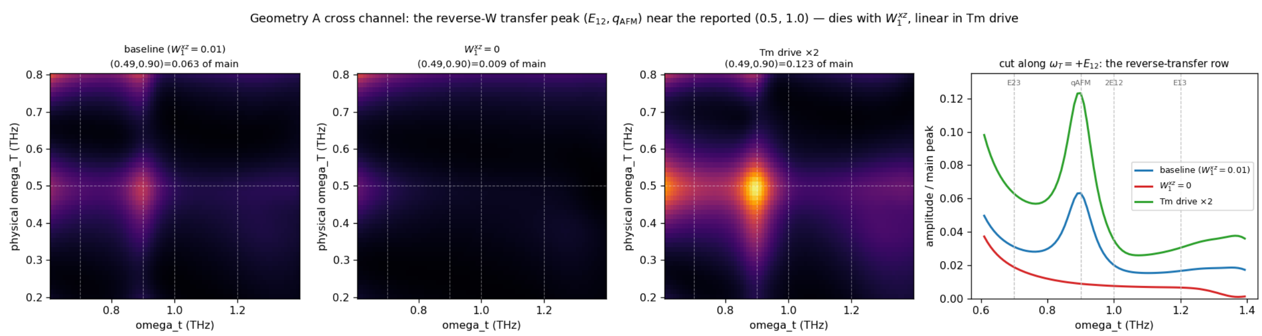

Figure S3: The reverse-W transfer peak in the geometry-A cross channel (λ^2), zoomed on the region of the reported (0.5, 1.0) feature; all amplitudes normalized to the main (q_{AFM}, E_{12}) peak. Left to right: baseline (genuine local maximum at $(+0.49, 0.90) = 0.063$), $W_1^{xz} = 0$ (collapsed to 0.009), Tm drive $\times 2$ (doubled to 0.123). Rightmost: the $\omega_T = +E_{12}$ row cut—the peak sits at the q_{AFM} emission frequency with the $2E_{12} = 1.00$ THz harmonic as a shoulder.

The population channel demonstrated. To make the T_1 scenario concrete we ran the geometry-B single-pulse protocol with the population channel switched on ($W_3^{yz}\, S_y S_z \lambda^3 = 0.01$, $\gamma_{\lambda^3} = 0.02$, i.e. $T_1 = 33$ ps; runs `bb3_W3` versus `bb3_ctl`). The pulse deposits Δn with $\Delta\lambda^3 = -4.6 \times 10^{-4}$, which then relaxes on the set T_1 exactly ($-3.6 \to -2.0 \to -0.8 \to -0.2 \times 10^{-4}$ at 13/33/66/132 ps); the q_{AFM} *oscillation* is bit-identical with and without W_3 (populations do not oscillate), and the population-fed quasi-static background scales as $W_3\, \Delta n$—of order 10^{-7} here, far below the coherent signals, because *coherent* THz pumping only produces $\Delta n \sim 5 \times 10^{-4}$. The quantitative conclusion: a sizable experimental post-pulse buildup requires *incoherent* population transfer (absorption $\to$ phonons $\to$ Tm repopulation, the mechanism of thermally driven spin reorientation), with amplitude proportional to $\Delta n(t)$ and growth time equal to the repopulation time—both measurable, neither present in a purely coherent drive model.

The population grating also fails—for a sharp reason. One might hope the τ-labeled population created jointly by pump and probe (the "population grating", which carries the E_{12} delay phase and survives on T_1) could feed a delayed, phase-labeled magnon signal. A full 2DCS run with the population channel on ($W_3^{yz} = 0.01$, $T_1 = 100$ code units; run `bb4_W3grating`) shows the $\omega_\tau = E_{12}$ row *identical* to the W_3-off baseline at every detection time. The reason is structural: the grating's delay label is written by the E_{12} *coherence* before the probe arrives, so its τ-support is confined to $|\tau| \lesssim T_2(E_{12})$, and its W_3 force is a step at the probe—an impulsive launch, not a slow drive. No element of the coherent model class produces post-pulse growth.

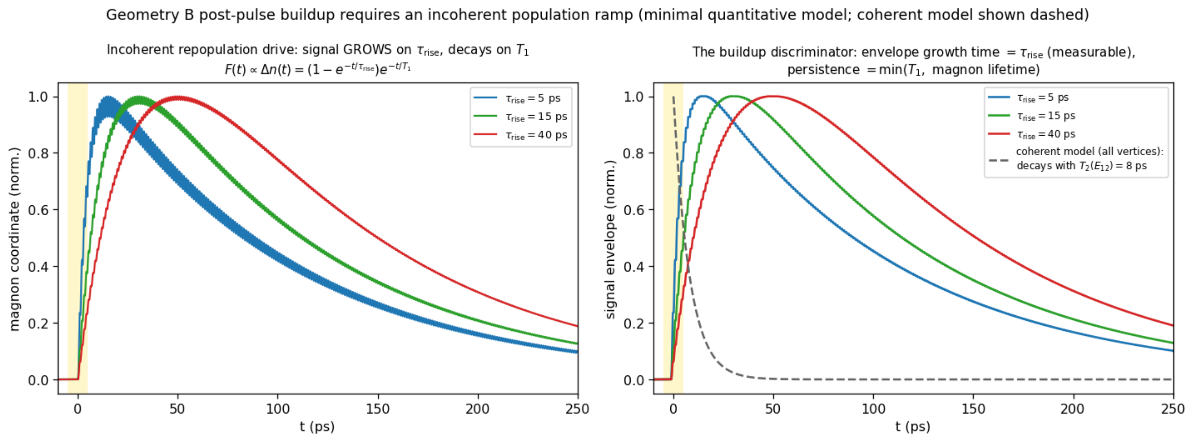

Figure S4: Minimal quantitative model of the observed buildup: a magnon (measured frequency and linewidth) driven by an *incoherent* population ramp $F(t) \propto \Delta n(t) = (1 - e^{-t/\tau_{\rm rise}})\, e^{-t/T_1}$. The signal grows on $\tau_{\rm rise}$, peaks at tens of ps, and persists on $\min(T_1, \text{magnon lifetime})$—the experimental phenomenology—whereas the full coherent model (dashed) decays with $T_2(E_{12}) = 8\,$ps. Fitting the experimental envelope to this form measures $\tau_{\rm rise}$ (the phonon-mediated Tm repopulation time) and the product $W_3\, \Delta n$.

The mechanism implemented in the solver. The incoherent channel is now part of the simulation: the solver accumulates the *dissipated* coherence energy, $E_{\rm dep}(t) = \int dt' \sum_{i,a\neq 3,8} \Gamma_a \big(\lambda_i^a - \lambda_{i,\rm eq}^a\big)^2$ (the energy the bath actually receives; the population channels are excluded from the sum, which closes an otherwise runaway self-heating loop), and shifts the λ^3 equilibrium by $-c_{\rm heat} E_{\rm dep}$. The existing Γ_3 relaxation then chases the shifted equilibrium, so the repopulation rise time is $1/\Gamma_3$ by construction, and the pump-FID×probe interference contained in the dissipated energy carries the $\omega_\tau = E_{12}$ label through the $M_{\rm NL}$ subtraction automatically. A full 2DCS run with $W_3^{yz} = 0.01$, $1/\Gamma_3 = 33\,$ps, and $c_{\rm heat}$ set for a few-percent population shift (run `bb5c`) reproduces the observed phenomenology [Fig. S5]: the $\omega_\tau = E_{12}$ row falls off the impulsive peak, *builds back up* on the repopulation time, and persists beyond 100 ps—50–100× above the coherent-only model at late times. The two new parameters map onto measurables: the recovery time of the row is $1/\Gamma_3$, and the late-to-impulsive intensity ratio calibrates $c_{\rm heat}\, W_3$.

If the experimental buildup over tens of ps is real, it must be fed by the one reservoir that outlives every T_2: the CEF *populations*. A pulse-deposited excited population n_2 relaxes on T_1, and through the population channel of the same exchange family ($W_3\, S_\alpha S_\beta\, \lambda^3$, implemented in the solver but not part of the present minimal Hamiltonian) it shifts the effective Fe anisotropy as it relaxes—precisely the mechanism of thermally driven spin reorientation in $TmFeO_3$. This route makes a sharp experimental discriminator: a population-fed buildup is a *non-oscillatory* background (or a slow shift of the magnon frequency/equilibrium) whose growth time directly measures T_1, whereas growth of the 0.90 THz *oscillation amplitude* over tens of ps would require a persistent coherent source that the measured T_2's exclude—observing it would falsify the linewidth-constrained model class itself.

Why the static vertex cannot replace the gated one. $W_1^{xz} S_x S_z \lambda^1$ is trilinear; around the Γ_2 order it decomposes into a static CEF shift, two energy-conserving *bilinear* hops ($F_x\, \delta S_z\, \lambda^1$ and $G_z\, \delta S_x\, \lambda^1$: pure hybridization), and a genuine two-fluctuation converter $\delta S_x\, \delta S_z\, \lambda^1$. The converter piece is real and retained—it drives the post-pulse buildup through the nearly closed triad $q_{\rm FM} + E_{12} = 0.88 \approx q_{\rm AFM}$ (0.02 THz detuning)—but it is fluctuation-fed (its energy leg is a probe-

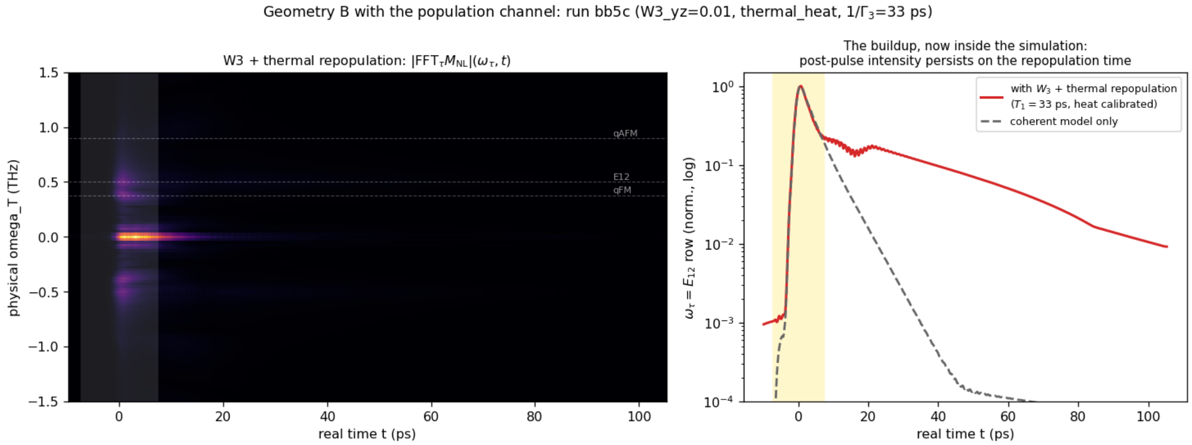

Figure S5: The buildup inside the simulation (geometry B, run `bb5c`: $W_3^{yz} = 0.01$, dissipation-fed thermal repopulation, $1/\Gamma_3 = 33\,\mathrm{ps}$). Left: mixed-domain map $|\mathrm{FFT}_\tau M_{\mathrm{NL}}|(\omega_\tau, t)$. Right: the $\omega_\tau = E_{12}$ row—after the impulsive peak the signal recovers on the repopulation time and persists beyond 100 ps (red), versus the coherent-only model dying by 40 ps (dashed).

launched magnon, one factor of drive×susceptibility weaker and spectrally narrow) and detuning-limited (secular accumulation over tens of ps), so it produces an *envelope*, not an impulsive phase-stamped kick. At $W_1^{xz} = 0.01$ the isolated run has sizable absolute amplitude at $(+0.5, 0.90)$ (0.72 of map max) but it is the *tail* of the magnon–magnon peak—no genuine local maximum on the E_{12} row.

A careful strength sweep in the consolidated configuration ($W_1^{xz} \in \{0.01, 0.02, 0.03, 0.05, 0.07, 0.09\}$, $\kappa^B = 0$; runs `vBwxz*`) shows that raising the static vertex *never* creates a second resolved peak. Instead the two delay rows coalesce—the leading peak's delay phase migrates continuously, $0.38 \to 0.40 \to 0.48 \to 0.47$ THz across the sweep, the resolved $(q_{\mathrm{FM}}, q_{\mathrm{AFM}})/(E_{12}, q_{\mathrm{AFM}})$ doublet structure never appears at any strength, and the hybridization-shifted E_{12} branch leaks into the M1 channel as emission strays at $\omega_t = 0.44$–0.45 THz (growing to 0.15–0.16 by $W_1^{xz} = 0.09$). This is the structural signature of a static mixer: it *moves* peaks (level repulsion of the hybridized normal modes—also shifting the mode frequencies off their 1D-measured values), whereas the experiment shows two resolved peaks at the *unshifted* delay energies 0.38 and 0.50 THz. A gated converter *adds* a peak while leaving the normal modes untouched; with $\kappa_{1y}^B = 0.0035$ the census gives $(+0.38, 0.90) = 1.00$ and a separate $(+0.53, 0.89) = 0.46$, both on the measured lines. The necessity of the field-gated vertex therefore does not rest on any operating-point fine-tuning: static mixing and gated conversion are qualitatively different operations on the 2D map. In geometry B the Tm CEF channels λ^{4-7} are *exactly* zero (machine precision): the $E \parallel a, H \parallel c$ drive closes on the $\{\lambda^1, \lambda^2, \lambda^3\}$ subalgebra, an SU(2) closure that protects geometry B from CEF contamination.

Finite temperature (regenerated final-scenario runs `vB2`): $(q_{\mathrm{FM}}, q_{\mathrm{AFM}})$ is a pure two-magnon process and temperature-flat; $(E_{12}, q_{\mathrm{AFM}}) \propto n_1 - n_2$ falls $0.46\,(0\,\mathrm{K}) \to 0.45\,(5) \to 0.40\,(8) \to 0.36\,(10\,\mathrm{K})$. Both peaks remain resolved through ~ 10 K, defining the working window 5–8 K.

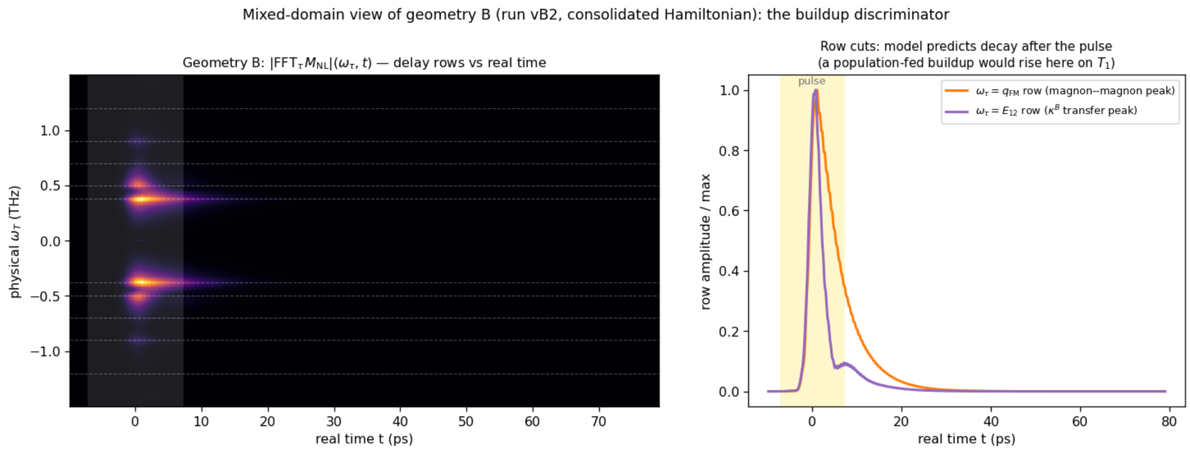

Figure S6: The buildup discriminator in the representation directly comparable to experiment: Fourier transform of $M_{\mathrm{NL}}(\tau, t)$ along the delay τ *only*, keeping the real detection time t (geometry B, run `vB2`). Left: $|\mathrm{FFT}_\tau|(\omega_\tau, t)$ map—each delay-frequency row evolves in real time. Right: cuts of the $\omega_\tau = q_{\mathrm{FM}}$ (magnon–magnon) and $\omega_\tau = E_{12}$ (κ^B-transfer) rows. The linewidth-constrained model predicts both rows peak with the pulse and decay within $\sim$20–30 ps; an experimental version of this plot in which the E_{12} row *rises* over tens of ps is the signature of the population-fed (T_1) mechanism.

S8 The same-polarization channel: pathways, exact single-ion verification, and the falsification ladder

S8.1 Configuration

The final same-polarization configuration (run `it14hr`; parameter files in `tmfeo3_2dcs_final/params_samepol_fina` is the unchanged v4 Hamiltonian with: measured pulse; drive vector $(0, 3.0, 0, 0, \mathbf{0}, 0, 0.9128, 0)$ on $(\lambda^1, \ldots, \lambda^8)$—i.e. the $1 \to 3$ leg removed ($\mu^z_{13} = 0$ in drive); per-leg amplitude twice the weak-probe baseline; dampings of Table S2 plus population relaxation $\gamma(\lambda^{3,8}) = 0.08$; Boltzmann ensemble; detection composite $0.05\,\lambda^4 + 4.4\,\lambda^6$ ($\mu^z_{13}/\mu^z_{23} \approx 1\%$ in emission), magnon M1 excluded per Table S2.

S8.2 Pathway assignments

With $\mu^z_{13} = 0$ the Liouville pathways of the five listed peaks are:

peak (ω_T, ω_t)	model (10 K)	side	pathway
$(+0.49, 0.72)$	1.00	NR	$\rho_{12} \xrightarrow{\text{probe } 1\leftrightarrow 2} n_2(\tau\text{-phase}) \xrightarrow{\text{probe } 2\leftrightarrow 3} \rho_{23}$
(0.5, 0)	sharpest S_{DC} line	—	same storage, zero-frequency output leg
$(+0.49, 1.20)$	0.78	NR	$\rho_{12} \xrightarrow{\text{probe } 2\leftrightarrow 3} \rho_{13}$, emission throttled by μ^z_{13}
$(+0.91, 1.18)$	$0.20 \to 0.51$ ($10 \to 20$ K)	NR	q_{AFM} delay, Fe–Tm handoff, thermally assisted
(0.9, 0.38)	absent	—	carrier exists (b-axis M1 q_{FM} ridge); no $q_{\mathrm{AFM}} \to q_{\mathrm{FM}}$ vertex
$(-0.47, 0.70)$	0.55	R	rephasing remnant (exact single-ion floor; prediction)
$(+0.99, 0.71)$	0.49	NR	inter-site $2E_{12}$ two-quantum line (prediction)

S8.3 Exact single-ion verification

A standalone integrator of the 8-dimensional $\vec{\lambda}$ equation (same CEF energies, dampings, measured pulse, same census pipeline; ~ 13 s per 2D map versus ~ 10 min for the lattice) reproduces the lattice 0.5-family census exactly, proving the family is single-ion physics and enabling exhaustive scans. Two facts emerge: (i) the rephasing/non-rephasing ratio of the (E_{12}, E_{23}) transfer peak has a floor of 0.47–0.58 under this pulse across all drive strengths, dipole-phase choices, and dampings—the rephasing remnant is an exact prediction of any three-level ion driven by this waveform; (ii) the $\omega_\tau = 2E_{12}$ two-quantum line is absent in the exact single-ion dynamics and is therefore an inter-site (mean-field) product.

S8.4 Falsification ladder

The dark-dipole scenario was reached by exhausting the alternatives; each row was run in the full lattice model with the blind census:

hypothesis	run	outcome
static mixing field $h_{\lambda^6} = 0.3$	it8	over-strong: transition lines shift off calibration
perturbative $h_{\lambda^6} = 0.06$	it9	pumps $(q_{\rm AFM}, E_{13})$ to 0.9 (wrong direction); side unchanged
CEF detuning 0.02 THz (ladder break)	it10	interloper survives; larger detuning barred by the observed degenerate lines
dipole triple-product sign flip	it11	census unchanged
dark μ_{13} at low drive	it12	population route floods the $(0, E_{23})$ band; needs the $2\times$ drive + T_1 combin
dark μ_{13}, $2\times$ drive, T_1	**it14**	**adopted**: both transfer peaks NR, correct hierarchy
negative μ_{13} leg, lower drive	it15	interloper returns ($\lvert\mu_{13}\rvert$ acts sign-independently); magnon-delay family floo

The $(0.9, 0.38)$ peak remains the one open item: the $q_{\rm FM}$ emission carrier exists as a uniformly τ-modulated b-axis M1 ridge in the Fe S_y channel (Γ_2 puts $\delta F_{y,z}$ at $q_{\rm FM}$), but the model lacks a $q_{\rm AFM} \to q_{\rm FM}$ transfer vertex of sufficient strength; one probed configuration (it15) produced a genuine $(+0.89, 0.33)$ maximum at 0.21 relative, evidence the peak is reachable in an extended Hamiltonian class.

S9 The consolidated configuration and its verified censuses

Every spectrum in the paper is produced by *one* Hamiltonian and *one* drive model; only the field geometry changes between channels. Couplings (code units): Fe exchange/anisotropy/DM of the calibrated orthoferrite model ($D_1 = 0.049$, $D_2 = 0.014$); Tm CEF $e_1 = 2.0678$, $e_2 = 4.9628$ meV; Fe–Tm $W_1^{yy} = 0.01$, $W_1^{xz} = 0.01$, $\kappa_{1y}^B = 0.0035$, population channel $W_3^{yz} = 0.01$ with the thermal-repopulation feedback ($\Gamma_3 = \Gamma_8 = 0.02$, i.e. $T_1 = 33$ ps; $c_{\rm heat} = 2000$, saturation cap 0.05, heat-reservoir cooling 1/300 code units); the λ-leg of every $SS\lambda$ vertex couples to the deviation $\lambda - \lambda_{\rm eq}$ (the static equilibrium contribution is already absorbed in the measured Fe parameters—without this renormalization the static $W_3\lambda^3$ piece shifts the magnon lines by $\sim$7%); drive legs $(d_z, \mu_{13}^z, \mu_{23}^z) \propto$ (3.0, 0, 0.9128) for $E \parallel c$ (the dark-μ_{13} rule) and the auto-projected Zeeman drive for the magnetic field; dampings and emission weights from Table S2. The final regeneration (runs `fin_A`, `fin_B`, `fin_S`; three Boltzmann seeds each, thermal channel included) gives the blind censuses, normalized per channel [amp, (ω_T, ω_t), $+ =$ non-rephasing]:

channel	leading genuine peaks (T=0)	reading
A cross (E1+M1 comp.)	1.00 $(+0.49, 0.90)$; 0.91 $(+0.90, 0.48)$; magnon diag. absent	W_1^{xz} reverse; W_1^{yy} forward
B cross (S_x)	1.00 $(+0.38, 0.90)$; 0.45 $(+0.53, 0.89)$; 0.17 $(-0.42, 0.90)$	magnon–magnon; κ^B transfer; R pa
A same (composite)	1.00 $(+0.49, 0.71)$[1]; 0.72 $(+0.49, 1.20)$; 0.59 $(-0.48, 0.70)$	population-route transfer; μ_{13}-thrott

The complete censuses at all temperatures, together with the normalized spectra themselves, are archived as `paper/data/census_final_picture.json`, `census_final_scenario.json`, and `paper/data/*.npz`; the buildup magnitude at this conservative calibration is a free dial (c_{heat}) that the measured late-to-impulsive intensity ratio will pin.

S10 Data and code availability

The paper folder is self-contained: `paper/params/` holds the exact `.param` files of every final-scenario run (`vA`, `vB2`, `it14hr`; three thermal seeds each), `paper/data/` the distilled spectra and censuses, `paper/scripts/` the census pipeline and the standalone qutrit integrator. The long-form derivation note (`tmfeo3_foundation.tex`: transported-gauge orbit formulas, vertex diagrams, complete written-out Hamiltonian, mode inventory) is archived alongside (`tfo_project/`); every spectrum in the paper can be regenerated from the committed `.param` files with the public `ClassicalSpin_Cpp` solver.

[1]Cross peaks normalized to the leading cross peak; the $\omega_T \approx 0$ pump–probe bands are excluded per the census convention of Sec. S4.